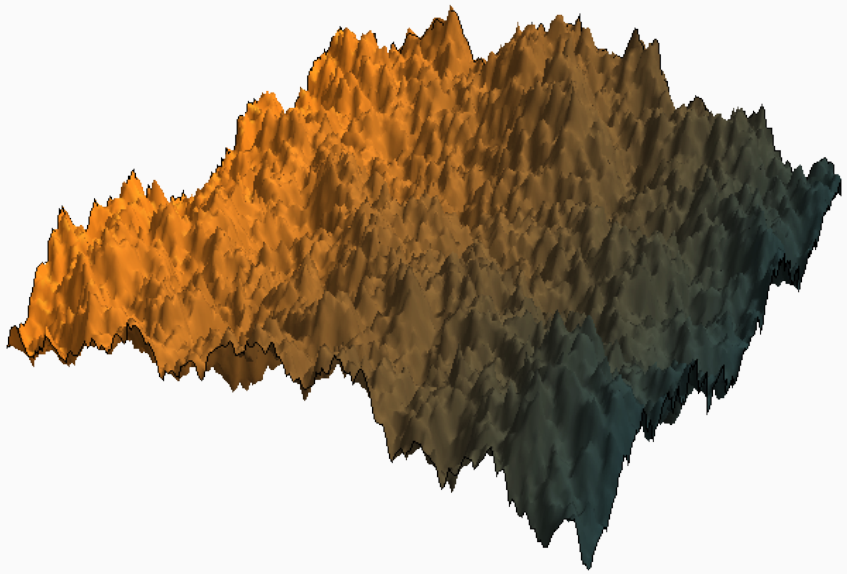
Quantum Field Theories
as
Statistical Field Theories



Antoine Tilloy
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Oberseminar Mathematische Physik LMU, Munich, Germany
February 1, 2017

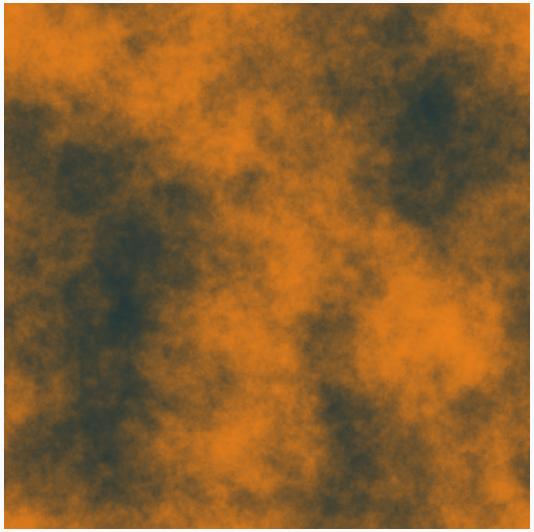
MPQ

First problem:

Quantum **field** theory is not about **fields**, there are no fields, it's about correlation functions of … macroscopic stuff made of …

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what we want

$$\langle \hat{\phi}(x_1)\hat{\phi}(x_2)\hat{\phi}(x_3)\hat{\phi}(x_4)\rangle = \sum \cdots$$

$$\langle \hat{\psi}(x_1)\hat{\psi}(x_2)\hat{\psi}(x_3)\hat{\psi}(x_4)\rangle = \sum \cdots$$

what we get

Second problem:

Even as operational theories, “physical” interacting quantum field theories are ill defined.

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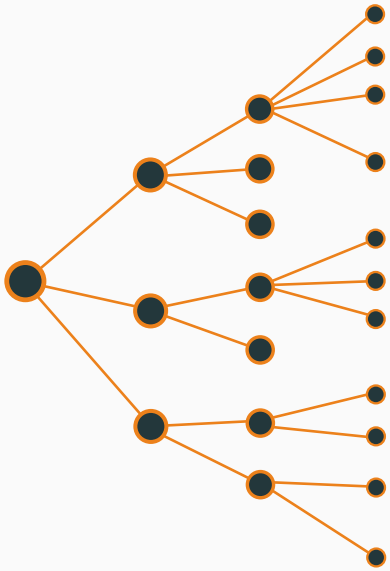
Even as operational theories, “physical” interacting quantum field theories are ill defined.

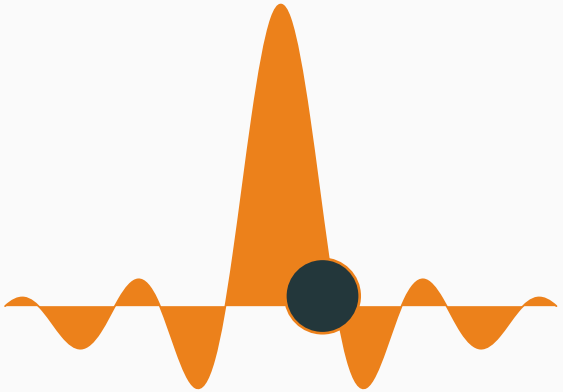
$$\langle \hat{\phi}(x_1)\hat{\phi}(x_2)\hat{\phi}(x_3)\hat{\phi}(x_4)\rangle = \sum_{k=0}^{+\infty} (\infty)_k \, g^k$$

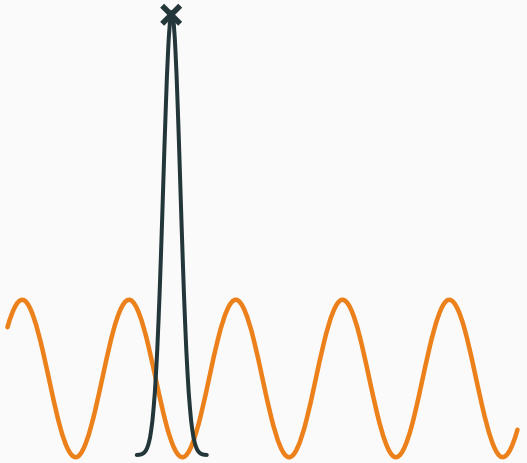
$$\langle \hat{\psi}(x_1)\hat{\psi}(x_2)\hat{\psi}(x_3)\hat{\psi}(x_4)\rangle = \sum_{k=0}^{+\infty} (\infty)_k \, g^k$$

These are actually the two classes of difficulties noted by Paul Dirac in 1963

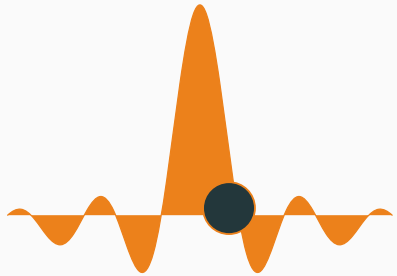
Reformulations of quantum theory are needed



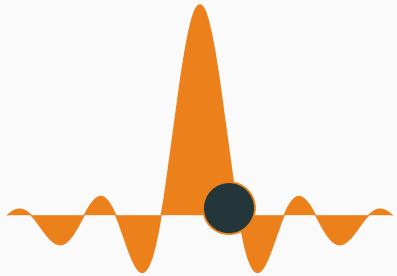




The Bohmian way

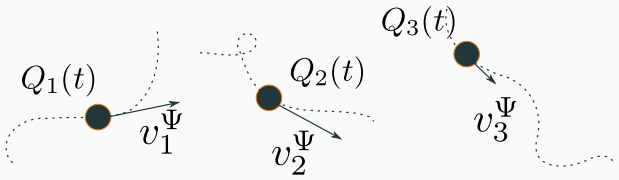


The Bohmian way



$$\frac{\mathrm{d}Q_k(t)}{\mathrm{d}t} = v_k^{\Psi}(Q_1(t), \cdots, Q_n(t))$$

$$i\hbar \frac{\mathrm{d}\Psi}{\mathrm{d}t} = H\Psi$$





L. de Broglie 1927



D. Bohm 1952

Subtleties

Lorentz invariance

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Either fixed foliation of foliation determined by the wave function $\rightarrow$ the letter but not the spirit

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Particle creation-annihilation

Subtleties

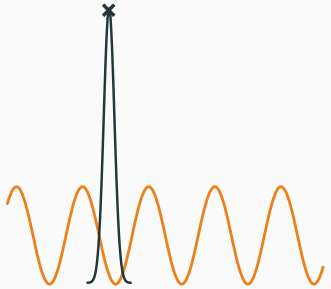
Lorentz invariance

Either fixed foliation of foliation determined by the wave function → the letter but not the spirit

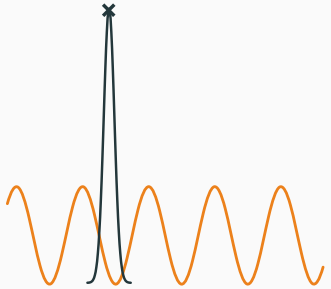
Particle creation-annihilation 2 solutions

- Stochastic creation-annihilation events
- Resurrect the Dirac sea

The Collapse way



The Collapse way



A modified Schrödinger equation

$$\partial_t |\psi_w\rangle = -iH|\psi_w\rangle + \mathrm{tiny}(w, \psi_w)$$

A primitive ontology

fields, particles, flashes

What is known:

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A Lorentz invariant GRW (Tumulka, 2006)

but no interactions and a wave function formalism

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In both cases anyway, predictions $\neq$ QFT.

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> Construct a relativistic collapse model, naturally written as a Lorentz invariant statistical field theory, that can be embedded within an “orthodox” interacting quantum field theory.

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1. Collapse models can be made equivalent to quantum theory
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> Construct a relativistic collapse model, naturally written as a Lorentz invariant statistical field theory, that can be embedded within an “orthodox” interacting quantum field theory.

In provocative form

1. Collapse models can be made equivalent to quantum theory
2. Quantum field theories can be written as statistical field theories
3. The two previous points are equivalent

DYNAMICAL REDUCTION MODELS

A modified Schrödinger equation

$$\partial_t |\psi_w\rangle = -iH|\psi_w\rangle + \mathrm{tiny}(w, \psi_w)$$

A primitive ontology

fields, particles, flashes

Linear collapse equation:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left|\psi_w(t)\right\rangle = \left\{-i\hat{H}_0 + \sqrt{\lambda}\int_{\mathbb{R}^3}\mathrm{d}^3\mathbf{x}\;\hat{M}_\sigma(\mathbf{x})\,\underset{\text{white}}{w_t(\mathbf{x})} - \frac{\sqrt{\lambda}}{2}\hat{M}_\sigma^2(\mathbf{x})\right\}\left|\psi_w(t)\right\rangle,$$

$$\text{with}\quad \hat{M}_\sigma(\mathbf{x}) = \frac{1}{\sigma(2\pi)^{\frac{3}{2}}}\sum_{k=1}^{N}\int_{\mathbb{R}^3}\mathrm{d}\mathbf{y}_k\, m_k\, e^{-\frac{|\mathbf{x}-\mathbf{y}_k|^2}{2\sigma^2}}\left|\mathbf{y}_k\right\rangle\left\langle\mathbf{y}_k\right|,$$

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Master equation for $\rho_t = \mathbb{E}\left[\left|\psi_w(t)\right\rangle\left\langle\psi_w(t)\right|\right]$

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}(t) = -i\left[\hat{H}_0,\hat{\rho}(t)\right] - \frac{\lambda}{2}\int_{\mathbb{R}^3}\mathrm{d}^3\mathbf{x}\;\left[\hat{M}_\sigma(\mathbf{x}),\left[\hat{M}_\sigma(\mathbf{x}),\hat{\rho}(t)\right]\right].$$

AN INSTANTIATION: THE CSL MODEL

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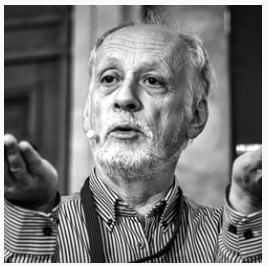
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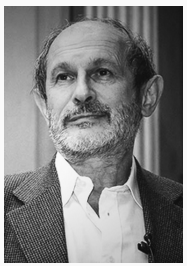
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More generally $\rho_t = \Phi_t\cdot\rho_0$
with Φ linear **C**ompletely **P**ositive **T**race **P**reserving

needs to be an “unraveling” of a nice open-system evolution



Gisin



Diosi

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Normalization and cooking:

$$\left|\widetilde{\psi}_w(t)\right\rangle = \frac{\left|\psi_w(t)\right\rangle}{\sqrt{\langle\psi_w(t)|\psi_w(t)\rangle}}$$

$$\mathrm{d}\mu_t(w) = \langle\psi_w(t)|\psi_w(t)\rangle\cdot\mathrm{d}\mu_0(w)$$

$$w_t(x) = 2\sqrt{\lambda}\,\langle\hat{M}_\sigma(x)\rangle + b_t(x).$$

white

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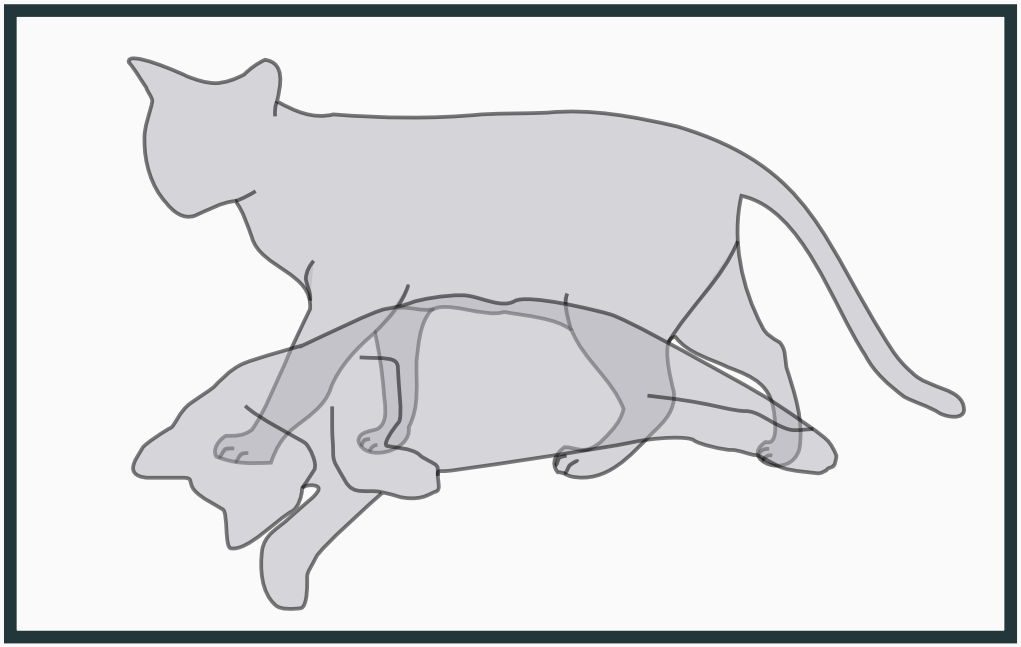
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A RECIPE FOR COLLAPSE MODELS

Start with a quantum system

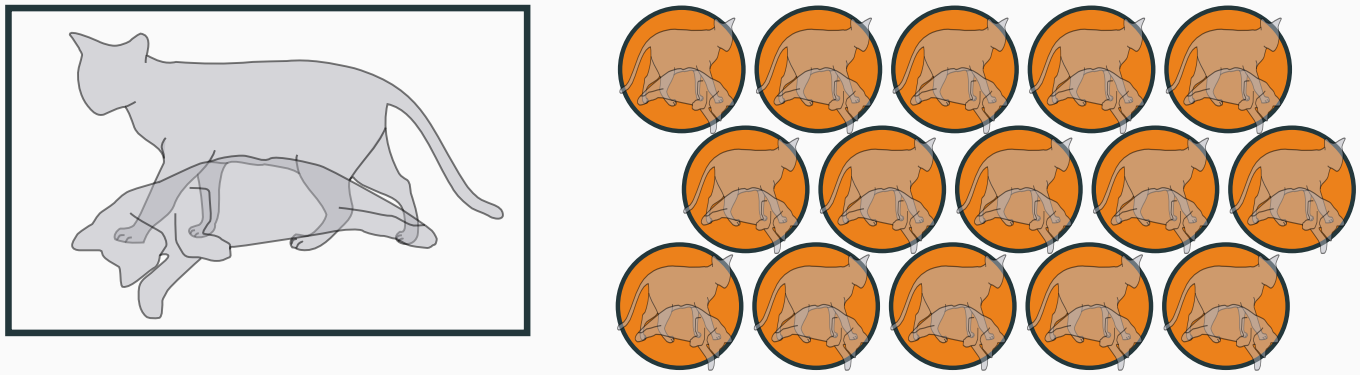
$$|\psi(t)\rangle$$



A RECIPE FOR COLLAPSE MODELS

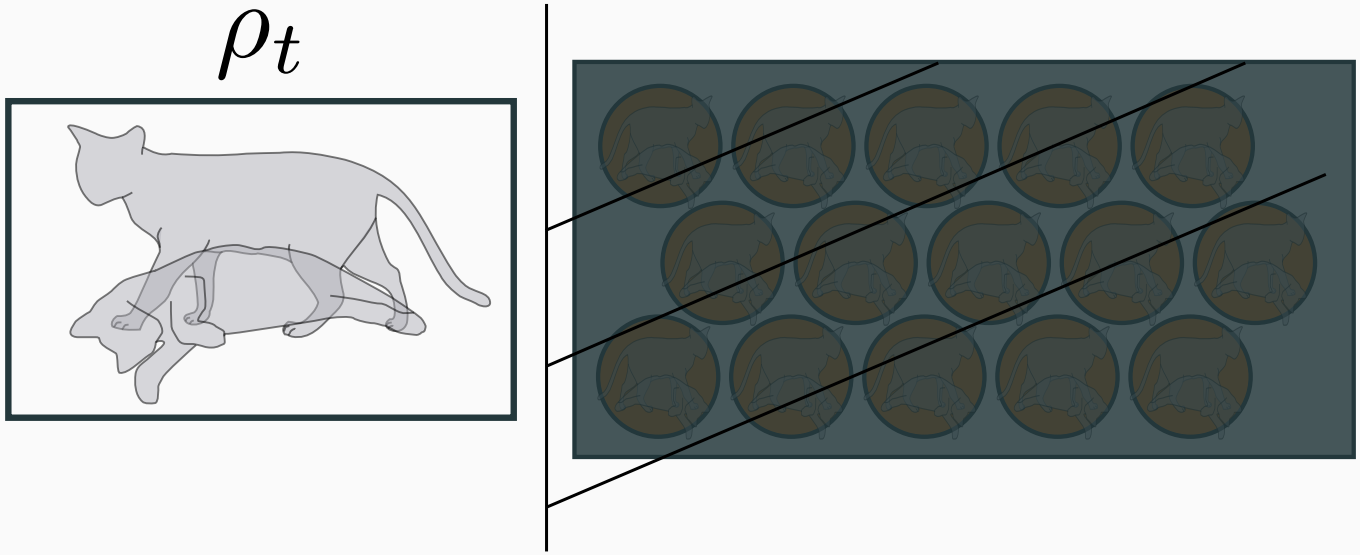
Couple it with a bosonic bath

$$|\psi(t)\rangle \otimes |\phi(t)\rangle$$



A RECIPE FOR COLLAPSE MODELS

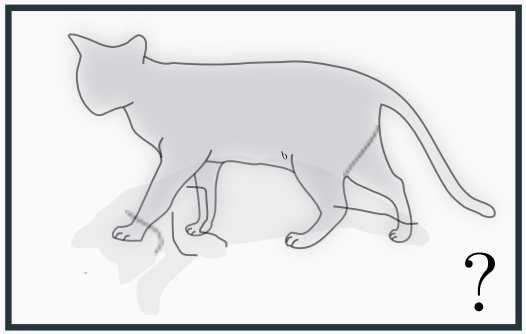
Trace out the bath



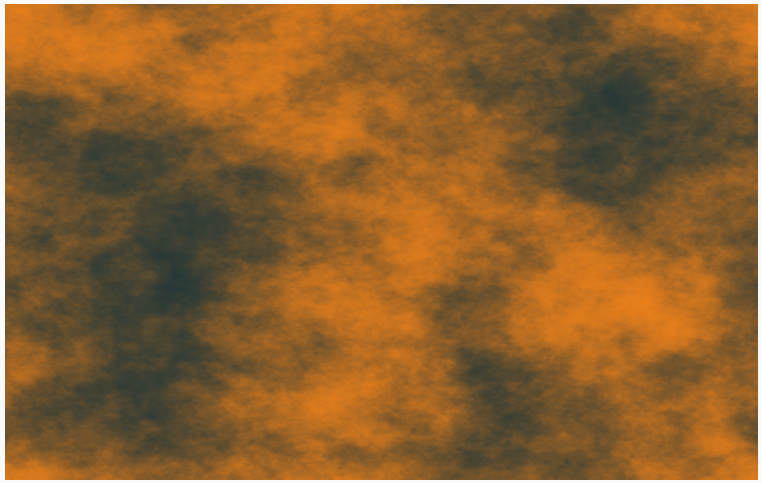
A RECIPE FOR COLLAPSE MODELS

Stochastically unravel the evolution

$|\psi_w(t)\rangle$



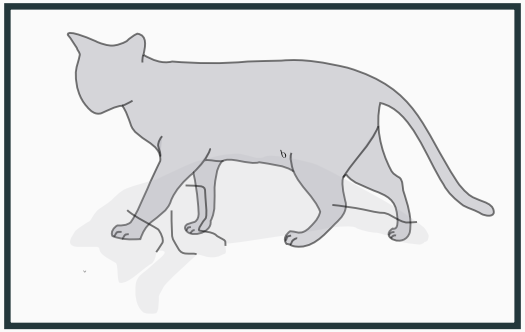
$\mathrm{d}\mu_0(w)$



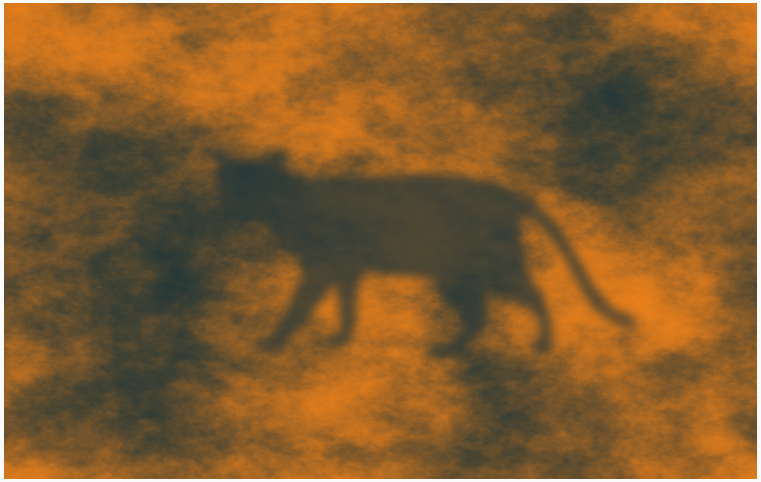
A RECIPE FOR COLLAPSE MODELS

Normalize the state and cook the measure

$$|\tilde{\psi}_w(t)\rangle$$

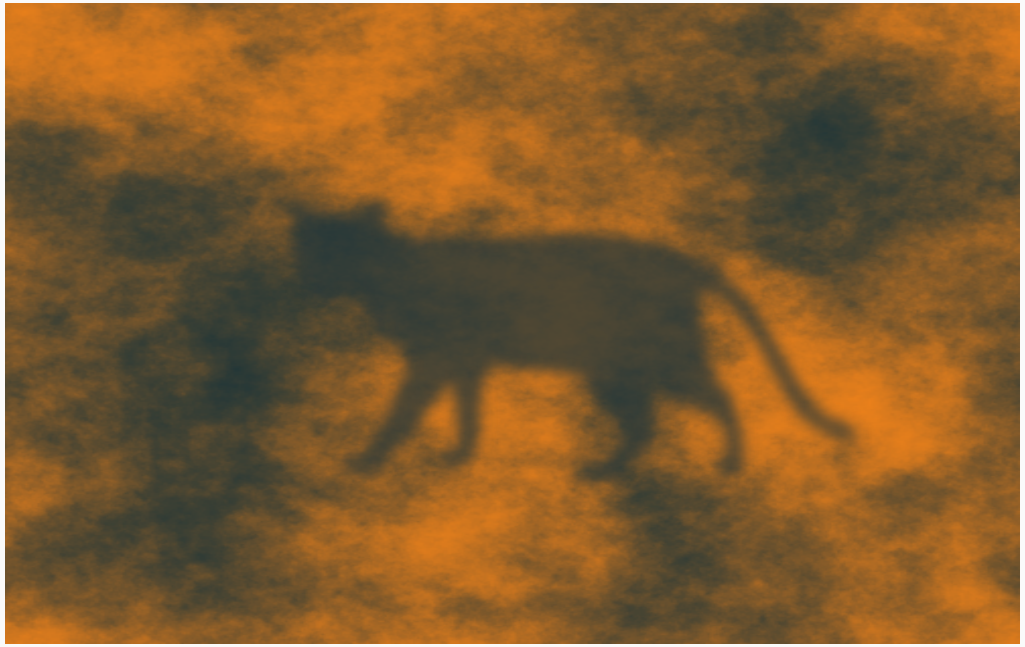


$$\mathrm{d}\mu_t(w)$$



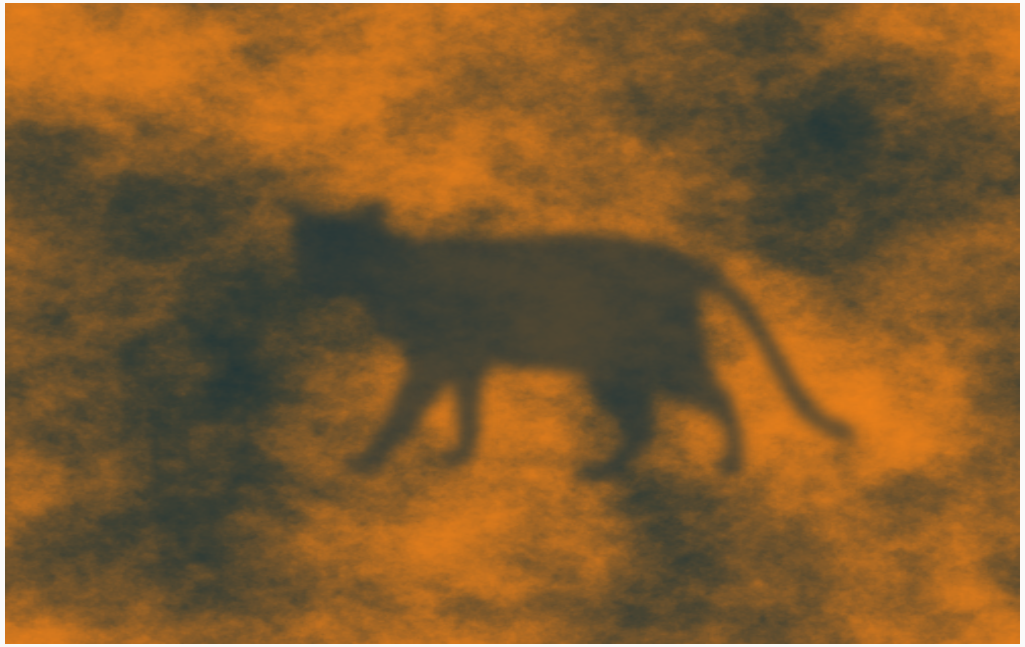
Keep only the primitive ontology

$$\mathrm{d}\mu_t(w)$$



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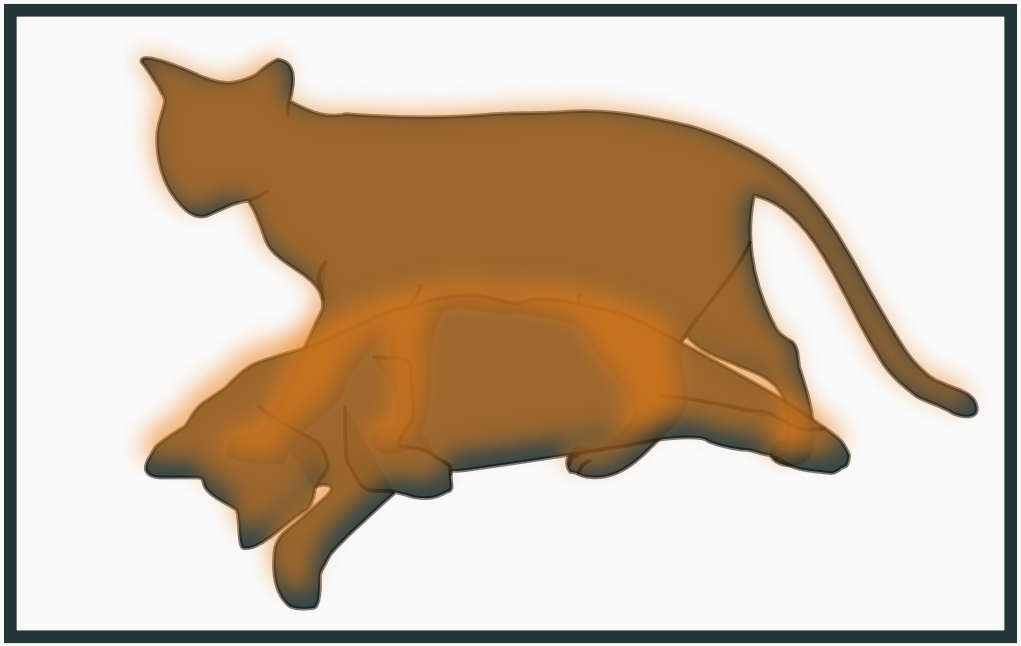


$\rightarrow$ you now have a single cat, living in $\mathbb{R}^3$

CHANGING THE RECIPE FOR QFT

Take an interacting QFT:

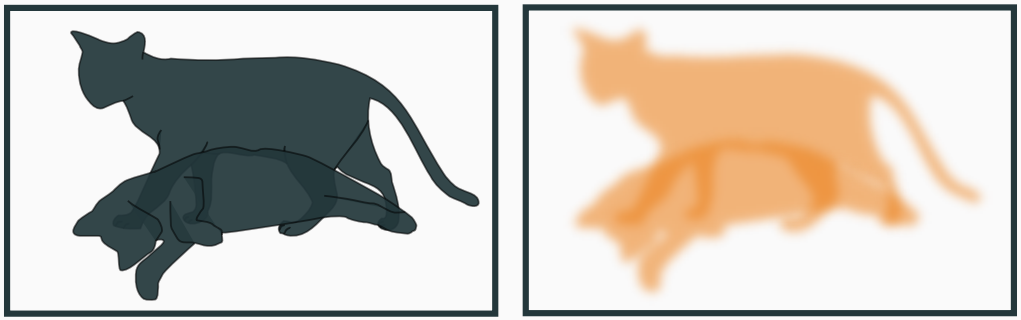
$$|\Psi(t)\rangle$$



CHANGING THE RECIPE FOR QFT

Split fermions and bosons:

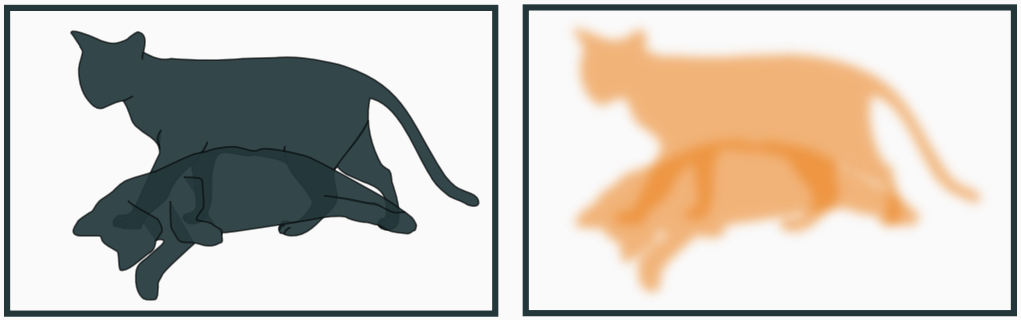
$$|\psi(t)\rangle \quad \otimes \quad |\phi(t)\rangle$$



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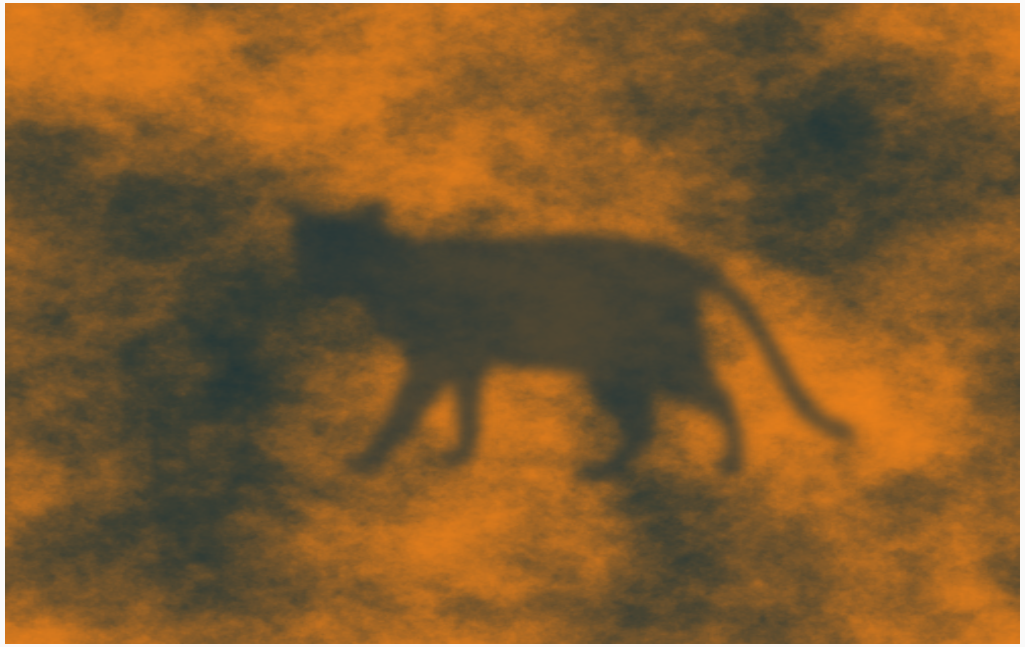
$$|\psi(t)\rangle \quad \otimes \quad |\phi(t)\rangle$$



$\rightarrow$ now do the same as before with the bosons as bath

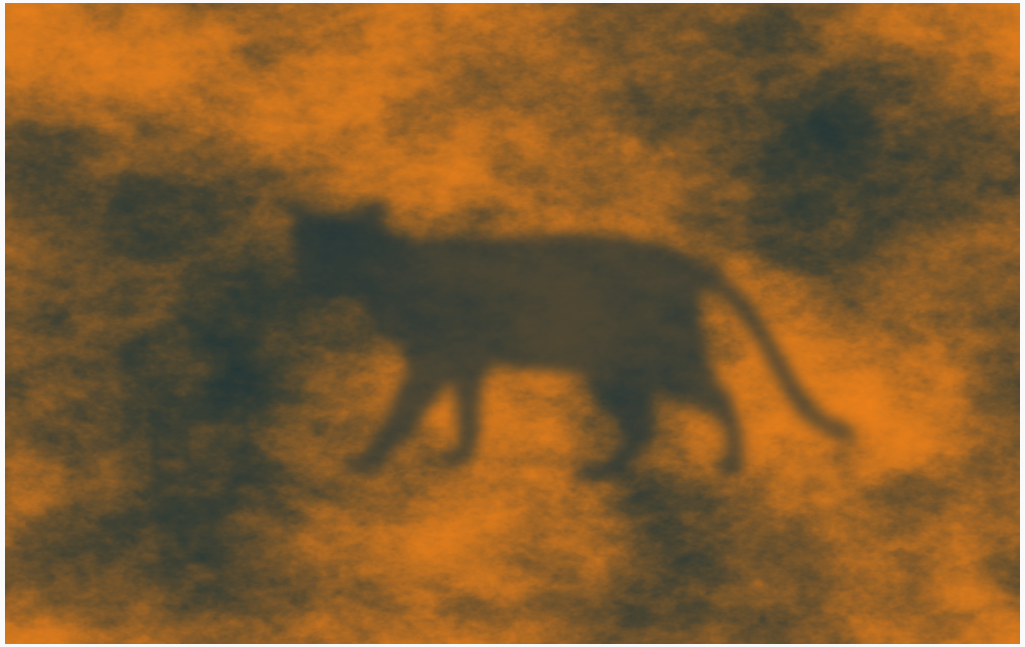
Final picture:

$$\mathrm{d}\mu_t(w)$$



Final picture:

$$\mathrm{d}\mu_t(w)$$



This time without changing the empirical content of the theory!

Consider a Yukawa theory:

$$\begin{aligned}
\mathscr{L}_f(\psi, \partial_\mu \psi) &= \bar{\psi}(i\partial\!\!\!/ - m_f)\psi \\
\mathscr{L}_{\text{int}}(\psi, \phi) &= g\,\bar{\psi}\,\psi\,\phi \\
\mathscr{L}_b(\phi, \partial_\mu \phi) &= \frac{1}{2}\partial_\mu \phi\, \partial^\mu \phi - \frac{1}{2} m_b^2 \phi^2,
\end{aligned}$$

IN PRACTICE

Consider a Yukawa theory:

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\end{aligned}$$

Take the momentum regularization Λ as fundamental

1) Integrating out the bosons

Define $\rho_t = \mathrm{tr}_b[\rho_{\mathrm{tot}}]$.

1) **Integrating out the bosons**

Define $\rho_t = \mathrm{tr}_b[\rho_{\mathrm{tot}}]$. Wick's theorem gives:

$$\hat{\rho}_f(t) = \mathcal{T} \exp\left(i\Phi\left[j^L, j^R\right]\right) \cdot \hat{\rho}_f(t_i)$$

where $\Phi[j^L, j^R]$ is the "operator" influence phase functional:

1) Integrating out the bosons

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$$i\Phi\left[j^L, j^R\right] = \int_{t_i}^{t}\int_{t_i}^{t} \mathrm{d}^4x\,\mathrm{d}^4y\; D(x,y)\, j^L(x) j^R(y)$$
$$- \frac{1}{2}\theta(x^0 - y^0)\, D(x,y)\, j^L(x) j^L(y) - \frac{1}{2}\theta(y^0 - x^0)\, D(x,y)\, j^R(x) j^R(y)$$

with $D(x,y) = \mathrm{tr}_b\left[\hat{\phi}(x)\hat{\phi}(y)\,\hat{\rho}_b(t_i)\right]$.

2) Unraveling the open evolution

$$
\begin{aligned}
|\psi_\xi(t)\rangle = &\mathcal{T}\exp\Bigg\{ -i\int_{t_i}^{t} \mathrm{d}^4x\, \hat{\jmath}(x)\,\xi(x) \\
&- \int_{t_i}^{t}\int_{t_i}^{t} \mathrm{d}^4x\,\mathrm{d}^4y\,\theta(x^0-y^0)\,[D-S](x,y)\,\hat{\jmath}(x)\hat{\jmath}(y)\Bigg\}\,|\psi_\xi(t)\rangle
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$$\mathbb{E}[\xi(x)\xi^*(y)] = D(x,y),$$
$$\mathbb{E}[\,\xi(x)\,\xi(y)\,] = S(x,y).$$

Note: importantly, finding ξ with the correlation matrix D is always possible (the relation matrix S is a free parameter)

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$$\mathrm{d}\mu_t(\xi) = \langle \psi_\xi(t) | \psi_\xi(t) \rangle \mathrm{d}\mu_{t_i}(\xi)$$

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In functional integral representation:

$$\mathrm{d}\mu_{+\infty} \propto \left| \int \mathcal{D}[\psi]\mathcal{D}[\bar{\psi}]\, \kappa[\bar{\psi}\psi] \exp\left(i\mathscr{S}_f + i\int \mathrm{d}^4x\, J\bar{\psi} + \bar{J}\psi - g\bar{\psi}\psi\xi \right) \right|^2 \mathrm{d}\mu_{-\infty}$$

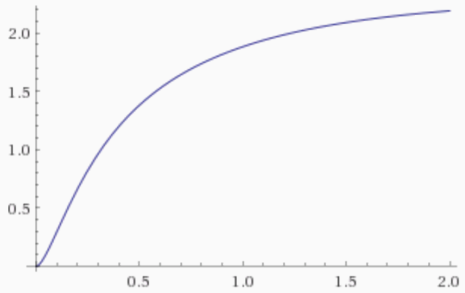
IN PRACTICE

One can prove **collapse** in position and **amplification**

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Collapse "strength" Ω between two points x and y:

$$\Omega_{+\infty}(\mathbf{x},\mathbf{y}) = \frac{-g^2}{(2\pi)^2}\left(K_0\left(m_b|\mathbf{x}-\mathbf{y}|\right) - K_0\left(\Lambda|\mathbf{x}-\mathbf{y}|\right) - \log\left[\frac{\Lambda}{m_b}\right]\right).$$



DISCUSSION

RECONSIDERING COLLAPSE

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- Collapse can be seen as an interpretation of QFT

RECONSIDERING COLLAPSE

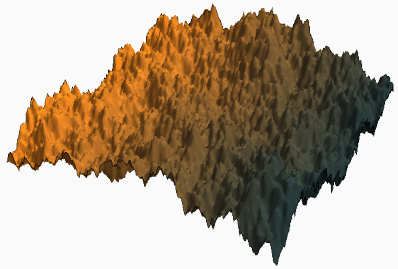
- Collapse can be seen as an interpretation of QFT
- Expected empirical signatures of collapse computed so far are artifacts of Markovianity / frame dependence.

RECONSIDERING QFT

1) QFTs can be seen as theories about fields

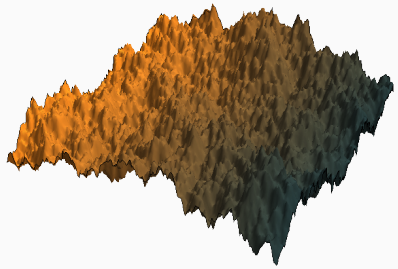
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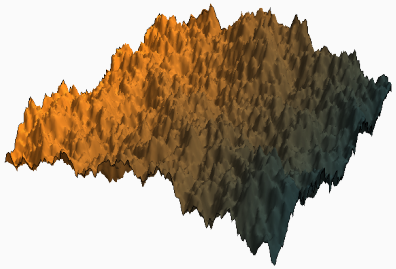
1) QFTs can be seen as theories about fields



2) QFTs can be fundamentally UV-regularized

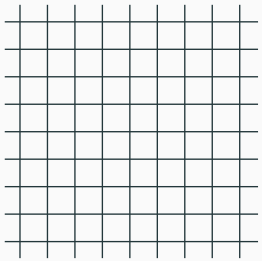
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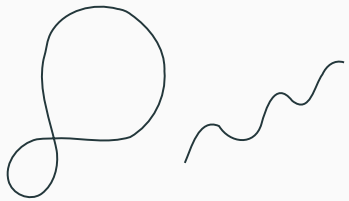
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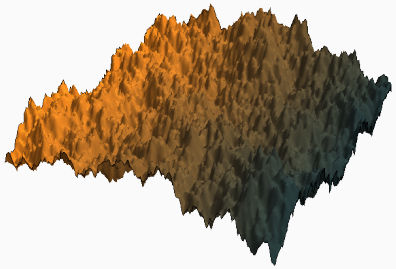
No Ostrogradsky instability, i.e. no need for:





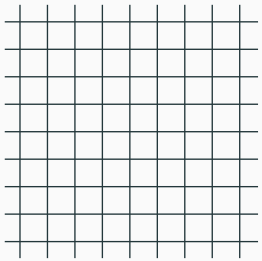
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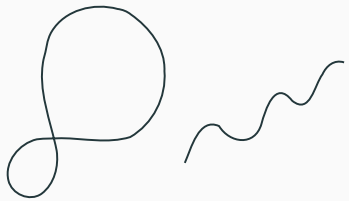
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→ makes Wilsonian renormalization transparent

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- It would be better to work from first principles

Summary

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Still analytical, numerical and conceptual cleaning to do